

Autonomous Traffic-Lightless Intersection Control

1. Stakeholder Identification

Entity / Stakeholder	Role in the system
Standard connected autonomous vehicles	Normal vehicles that approach the intersection, send driving information, receive instructions, and cross the intersection.
Emergency vehicles	Ambulances, fire trucks, or police vehicles that may require priority access through the intersection.
Intersection Coordination Unit / Edge Server	Central decision unit that collects vehicle information and computes safe crossing permissions or speed guidance.
Roadside Units (RSUs)	Infrastructure units near the intersection, including communication devices, cameras, and radars.
Vehicle sensors	Sensors inside vehicles used to estimate position, speed, acceleration, surrounding vehicles, and nearby obstacles.
Roadside cameras and radars	Infrastructure sensors used to detect traffic, obstacles, pedestrians, cyclists, and road conditions.
Pedestrians	Vulnerable road users who may cross the intersection and require protected crossing time.
Cyclists and e-scooter users	Vulnerable road users that may travel near the vehicle lanes or crossing area.
V2I / V2X communication network	Communication system that connects vehicles with infrastructure and supports information exchange.
Human passengers	Users inside vehicles; their safety, comfort, waiting time, and travel delay are important.
Traffic authority / city operator	Responsible for deployment, rules, monitoring, and legal operation of the intersection system.
Maintenance team	Responsible for repairing sensors, communication units, and computing infrastructure.
Non-connected or manually driven vehicles	Vehicles that may not follow automated instructions and create additional uncertainty.
Obstacles, animals, or road debris	Unexpected objects that can block the intersection or force vehicles to brake.

2. Problem Analysis

- The main problem is how to allow many vehicles from different directions to pass a traffic-lightless intersection safely, efficiently, and fairly. Traditional intersections use traffic lights or stop signs. These methods are simple and safe, but they often create unnecessary stopping, delay, congestion, idling, and reduced road capacity.
- In a traffic-lightless autonomous intersection, vehicles can pass with less stopping if their movements are coordinated correctly. However, this creates a difficult control problem because vehicles from different directions may reach the same conflict area at nearly the same time. The controller must therefore predict conflicts, reserve safe crossing space, and prevent collisions.
- The first paper focuses on global coordination of automated vehicles using an intersection coordination unit. Vehicles send their position, velocity, and destination, while the controller

computes safe trajectories by considering cross-collision risks at conflict points. The second paper focuses on robust autonomous intersection control using edge computing and V2I communication. It highlights that unstable communication channels can reduce safety, so vehicles must still behave safely even if they become disconnected.

3. System Constraints

Constraint	Explanation
Collision avoidance	No two vehicles should occupy the same conflict area or collision block at the same time.
Cross-collision point safety	Vehicles from different approaches must not reach the same collision point at the same time.
Speed limits	Vehicles must stay within allowed minimum and maximum velocity limits.
Acceleration and deceleration limits	Vehicles cannot accelerate or brake beyond their physical and safety limits.
Turning-speed constraint	Vehicles turning left or right must reduce speed for safe turning.
Minimum vehicle separation	Vehicles in the same lane must keep a safe distance from each other.
Communication reliability	The system should work safely even if V2I or V2X communication becomes unstable.
Sensor reliability	The system must handle missing, delayed, or inaccurate information from cameras, radars, or vehicle sensors.
Emergency handling	The system must react safely to unexpected obstacles, accidents, or sudden braking.
Fairness	No lane or vehicle should wait forever while other vehicles continue passing.
Traffic efficiency	The system should reduce waiting time and increase intersection capacity.
Computational limit	The controller must compute decisions quickly because intersection control is time-critical.
Vehicle autonomy	Vehicles must still be able to brake or stop locally if the central controller fails or sends no permission.
Priority rules	Emergency vehicles may need priority over normal traffic.

4. Operational Scenarios

Scenario 1: Normal 4-way traffic-lightless intersection

Vehicles approach from north, south, east, and west. Each vehicle sends its position, speed, destination, and movement intention to the controller. The controller decides the safe crossing order and gives permission or speed guidance.

Scenario 2: Vehicles going straight

Vehicles from opposite directions may pass at the same time if their paths do not conflict. Vehicles from crossing directions must be separated in time to avoid collision.

Scenario 3: Left-turning vehicle

A left-turning vehicle usually crosses more conflict areas. The controller must check all required conflict areas before giving permission. The vehicle must also slow down during the turn.

Scenario 4: Right-turning vehicle

A right-turning vehicle may have fewer conflict points, but it can still conflict with pedestrians, cyclists, or other vehicles. The controller must reserve the required path before allowing the turn.

Scenario 5: Emergency vehicle approaching

An ambulance, police car, or fire truck approaches the intersection. The system should give it priority by stopping or delaying normal vehicles in a safe way.

Scenario 6: Pedestrian crossing

A pedestrian enters or requests to cross. The system must reserve pedestrian crossing time and stop conflicting vehicle movements.

Scenario 7: Sensor failure

A roadside camera, radar, or vehicle sensor fails or gives unreliable data. The controller should switch to a safe mode, such as instructing vehicles to stop before the stop line until reliable data is available.

Scenario 8: Communication failure / disconnected vehicle

A vehicle loses connection with the edge server. If it already has permission, it follows the last safe instruction. If it has no permission, it must stop before the stop line.

Scenario 9: Obstacle or accident inside the intersection

An object, animal, broken vehicle, or accident blocks the intersection. The system must detect the blocked area, cancel unsafe permissions, and re-compute safe movements or stop vehicles.

Scenario 10: High traffic density / congestion

Many vehicles arrive from all directions at the same time. The controller must schedule vehicles efficiently, prevent starvation of any lane, and reduce waiting time.

Scenario 11: Mixed traffic with non-connected vehicles

A manually driven or non-connected vehicle enters the intersection area. The system should treat it as an uncertain object and increase safety margins. Connected vehicles should slow down or stop until the risk is clear.